CMP System Administrator's Manual

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§1. There are four parts to the computer music project: (1) The server; (2) The client machines; (3) The web services; (4) DISSCO. This manual will go through the general setup for the computer music project and the maintenance required by a system administrator.

Part I

Installation and Configuration

§2. The server's hostname is john.designcenter.illinois.edu with IP address 128.174.109.26.The client hostnames can be found on https://computermusic.web.illinois.edu/Notes/computers2.html.

§3. The CMP machines all have IP addresses between 128.174.109.0 and 128.174.109.255. This range can be achieved with a subnet mask 128.174.109.0/255.255.255.0.

§4. The server is running Debian 12 with Wayland/Gnome. The clients are running Debian 13 with X11/XFCE. The sudoer account for all systems is cmpadmin

§5. John runs NIS/NFS for managing client accounts, a pactl server for syncing audio, and a ufw firewall.

1 UFW Setup

§6. For all computers, be sure to ufw allow 128.174.109.0/255.255.255.0. Security is not too much of a concern since access to all computers requires being on the illinois intranet.

2 NFS setup

§7. NFS is setup using /etc/exports on the server and /etc/fstab on the clients.

§8 (**Server Setup**). On john, we export the home folder with the following line in /etc/exports.

```
/home    128.174.109.0/255.255.255.0(rw,sync,no_root_squash,subtree_check,insecure)
```

§9 (**Client Setup**). On the clients we mount john's /home/ folder with

```
128.174.109.26:/home     /home    nfs      fg,hard,_netdev 0       0
```

3 NIS Setup

Server Setup

§10. An NIS server is configured at /etc/hosts.allow, /etc/default/nis, /etc/yp.conf, /etc/ypserv.securenets, /etc/defaultdomain, /etc/nsswitch.conf, and /var/yp/-Makefile. Below will be what each of these files should be edited to. If a file is not listed then you can use the default configuration.

§11 (/etc/hosts.allow).

```
ALL : 128.174.109.19
ALL : 128.174.109.20
ALL : 128.174.109.21
ALL : 128.174.109.22
ALL : 128.174.109.23
ALL : 128.174.109.24
ALL : 128.174.109.25
ALL : 128.174.109.26
ALL : 128.174.109.27
ALL : 128.174.109.28
ALL : 128.174.109.29
ALL : 128.174.109.30
ALL : 128.174.109.31
```

§12 (/etc/yp.conf).

```
ypserver localhost
```

§13 (/etc/ypserv.securenets).

```
255.0.0.0       127.0.0.0
host            ::1
255.255.255.0   128.174.109.0
```

§14 (/etc/defaultdomain).

```
john.designcenter.illinois.edu
```

§15 (/etc/nsswitch.conf).

```
passwd:         files systemd nis
group:          files systemd nis
shadow:         files systemd nis
gshadow:        files systemd nis

hosts:          files myhostname mdns4_minimal [NOTFOUND=return] dns
networks:       files

protocols:      db files nis
services:       db files nis
ethers:         db files nis
rpc:            db files nis

netgroup:       nis
```

§16. Now, run "service ypbind restart" in order to restart the NIS service.

§17. Next, run "/usr/lib/yp/ypinit -m" in order to configure the NIS server.
Be sure john.music.illinois.edu is in the list.

§18. In /var/yp/Makefile you can change "MINGID=1000" to "MINGID=1"
to allow sudoers to be exported.

Client Setup

§19 (nsswitch.conf).

```
passwd:         files systemd nis
group:          files systemd nis
shadow:         files systemd nis
gshadow:        files systemd nis

hosts:          files myhostname mdns4_minimal [NOTFOUND=return] dns
networks:       files

protocols:      db files nis
services:       db files nis
ethers:         db files nis
rpc:            db files nis

netgroup:       nis
```

§20 (/etc/defaultdomain).

```
john.designcenter.illinois.edu
```

§21 (/etc/yp.conf).

```
ypserver 128.174.109.26
```

§22. To finish the NIS setup you must enable and start ypbind.service

4 Pactl Setup

§23 (**Server Setup**). On john, run "pactl load-module module-native-protocol-tcp listen=128.174.109.26" and then run "pactl load-module module-zeroconf-publish"

§24. Sometimes you may need to run these commands again if the audio server fails. You can make these commands run on login by editing /etc/pulse/default.pa.

§25 (**Client Setup**). Run "pactl load-module module-zeroconf-discover" to set up the client audio servers.

5 Installing a New Client Server from Scratch

§26. When going through the Debian installation process, make sure DHCP picks up the correct domain name for the computer. Select xfce as the desktop environment.

§27. To start with, install nis and nfs-common. You may also want to install vim, gedit and fish to make the next steps easier.

§28. Next run the NFS client setup as listed in § 9 on page 2. Then run the NIS client setup in section 3 on the previous page. Finally install pipewire and run the pactl client setup in § 25.

§29. Next install the necessary packages. I have a list of packages to install on /home/cmpadmin/programbundle. In the future this step should be automated with ansible.

The programs are: "apt install libgtkmm-2.4-dev audacity lilypond libxerces-c-dev libsndfile1-dev premake4 libmuparserx4.0.12 pipewire libmuparserx-dev libgtkmm-2.4-1v5 libmuparser-dev libmuparser2v5 fish vim git gedit emacs alpine zsh neovim micro pulseaudio-utils ssh openssh-server nis nfs-common gdb pipewire-pulse"

In /home/cmp/Downloads is the .deb file for vscode, this should be installed for the SPIN students.

§30. You need to enable and start ssh.service so that the computer can be used remotely.

§31. Sometimes Sever's account may not work locally, to fix this, simply use gdm3 instead of lightdm. (After using gdm3 once for sever, you can switch back to lightdm and it will work correctly.)

§32. It is useful to disable sleep and screen blanking to keep the computers awake. Run "sudo systemctl mask sleep.target suspend.target hibernate.target and hybrid-sleep.target".

You may need to edit /etc/gdm3/greeter.dconf-defaults by changing all of the "suspend" lines to "nothing" as well.

Part II

Maintenance

§33. Most of the maintanence relates to adding new users and running backups. You may also need to help students and rerun the pactl commands in § 25 on the previous page if the audio server stops working.

6 Adding new users

§34. New users should be added on john. The easiest way to do so is to use the command "newusers". The "newusers" command takes as input a text file, every line corresponding to a new user. The order goes

```
(username):(password):(user id):(group id):(Full name/misc info):(home dir):(shell)
```

The absolute minimum for a new user is the following

```
netid:password::::/home/netid:/bin/bash
```

§35. After making the new user on john, you must run "sudo make -C /var/yp" in order to sync the new user accross computers. It is useful to test logging in as one of the users you've created.

§36. As soon as a new user logs on, it is imperative that they change their password immediately using the command "yppasswd".

7 Backups

§37. We use borgbackup to run backups. I'd recommend doing this once a week. If you want you can set up a cron job to do this, but it's often useful to consolidate the backing up to the time where you check the server once a week. The command to create a backup is below, but be sure to change the backup location to wherever the external drive is mounted and "20260205" to whatever the current date is.

```
sudo borg create -v --stats /media/cmpadmin/Backup/borgbackup/::20260205 /home/
```

§38. Dissco files are just XML files, and so can be backed up with git very easily. I have set up a git repository for sever's compositions, but it's currently a bit difficult to use. It would be useful to automate these git backups on sever's computer so his compositions and their edit history are automatically saved.

8 Web Services

§39. The websites are hosted by tech services since the university does not want us to run publically accessible servers.

§40. These websites can be edited on https://web.illinois.edu/. If you want to add stuff to the website, Sever can add you to the cpanel.

Part III

General Tips

§41. Remember you need to be logged into the illnois vpn to access the computers. You can use the cisco service but it is easier to use openvpn.

§42. If the client computers freeze, one thing to do is to unplug the ethernet switches, wait for 30 seconds and plug them back in. This will fix the computers.

§43. The list of previous system administrators is, from earliest to latest is:

```
Ed White (??? - Spring 2001)
Nasko Oskov
Steve Hanna
Andy Ying
Ming-ching Chiu
Benjamin D. Whiting
```

Zhaoyu Zhang
M.O. Abbott
Dongryul Lee
???
Advith Govindarajan

Feel free to ask any of them if you need any help.